Mechanical Engineering — Semester II

Engineering Mechanics Lab

Course Code: 2029 | Credits: 1.5 | Contact Hours: 45 | Curriculum: Rev 2026

Download Lab Manual

Official Source Declaration

This lesson is architected based on the **Official Kerala Polytechnic Syllabus Revision 2026** for the course **Engineering Mechanics Lab (2029)**. All course objectives, outcomes, experiment lists, and assessment schemes are derived directly from the official document.

Note: Experimental procedures provided are standard engineering practices aligned with the syllabus requirements.

Course Dashboard

Teaching Scheme

- **Practical (P):** 3 Hours/Week
- **Self-Learning (SS):** 4.5 Hours/Week
- **Total Hours:** 45 Hours/Semester

Assessment Scheme

- **CIE (Internal):** 60 Marks
- **ESE (External):** 40 Marks
- **Total:** 100 Marks

How to Study This Lab Course

Engineering Mechanics Lab is about visualizing forces. To succeed:

1. **Pre-Lab:** Review the theoretical concepts (Parallelogram law, Lami's theorem, Friction) before entering the lab.
2. **Observation:** Record readings precisely. Small errors in reading spring balances or scales can lead to large errors in results.
3. **Graphical Verification:** Many experiments require comparing analytical results with graphical methods (like the Link Polygon).
4. **Design Thinking:** Module IV requires a shift from "following steps" to "solving problems."

Course Outcomes (CO)

CO	Description	Bloom's Level
CO1	Identify force systems for given conditions by applying basics of mechanics.	Apply
CO2	Determine unknown forces of different engineering systems.	Apply
CO3	Determine centre of gravity, mass moment of inertia and coefficient of friction.	Apply
CO4	Apply design thinking in the engineering domain.	Apply

Syllabus Coverage Map

Module-wise Experiment Distribution

Module	Focus Area	Experiments
I	Force Systems & Equilibrium	Expt 1 - 4
II	Beam Reactions	Expt 5 - 6
III	Properties of Surfaces & Friction	Expt 7 - 11
IV	Design Thinking Application	Expt 12 - 14

Learning Roadmap

Foundation: Understanding Concurrent Forces (Module I) → **Application:** Analyzing Structural Elements/Beams (Module II) → **Advanced Analysis:** Dynamic properties and surface interactions (Module III) → **Innovation:** Solving real-world problems through Design Thinking (Module IV).

Module I: Force Systems & Equilibrium

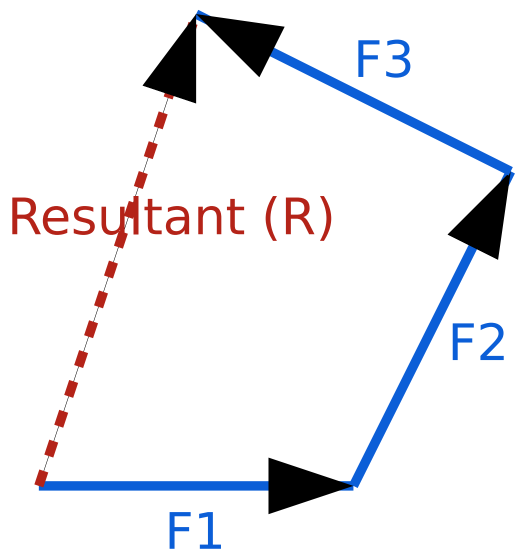
14 Instructional Hours | CO1

1. Concurrent Force Systems

When multiple forces act on a single point, they are called concurrent forces. In this module, we verify the laws that govern their resultant.

Experiment 2: Law of Polygon of Forces

Principle: If any number of forces acting at a point can be represented in magnitude and direction by the sides of a polygon taken in order, then their resultant is represented by the closing side of the polygon taken in opposite order.



Vector Polygon Construction for Resultant Determination

Malayalam support:

[illegible] (Forces) [illegible] (Resultant) [illegible] Polygon Law [illegible]. [illegible] (Polygon) [illegible], [illegible] Resultant.

Experiment 4: Jib Crane & Lami's Theorem

Lami's Theorem: If three coplanar forces acting at a point are in equilibrium, each force is proportional to the sine of the angle between the other two forces.

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P / sin(α) = Q / sin(β) = R / sin(γ)
```

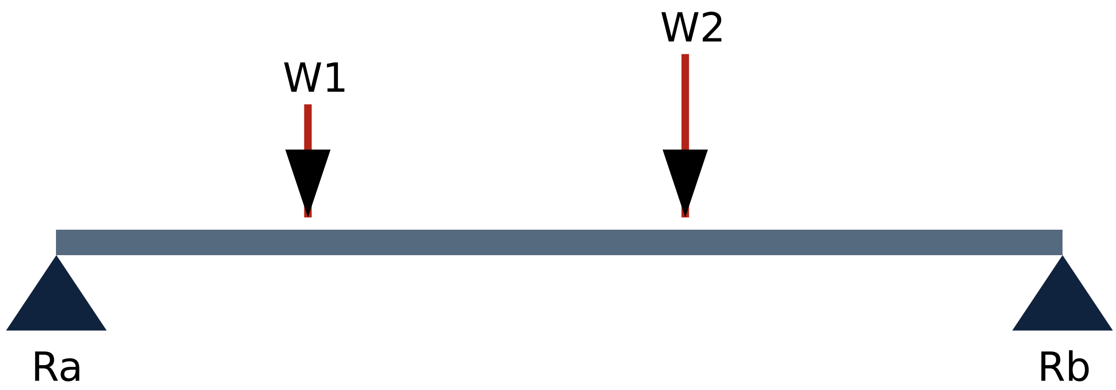
Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module II: Support Reactions of Beams

6 Instructional Hours | CO2

Experiment 5 & 6: Simply Supported Beams

Beams are structural members that primarily resist loads applied laterally to their axis. In a simply supported beam, the supports are at the ends.



Simply Supported Beam with Point Loads

Example Calculation:
Given: Beam length = 4m. Load W1 = 100N at 1m from left support (A).
Required: Reactions Ra and Rb.
Calculation:
ΣMa = 0 (Taking moments about A):
(100N × 1m) - (Rb × 4m) = 0
100 = 4Rb => **Rb = 25N**
ΣFy = 0 (Vertical forces):
Ra + Rb = 100N

Ra + 25 = 100 => **Ra = 75N**

Answer: Ra = 75N, Rb = 25N.

Module III: Centroid, Inertia & Friction

15 Instructional Hours | CO3

Friction (Expt 10 & 11)

Friction is the force resisting the relative motion of solid surfaces. The coefficient of friction (μ) is the ratio of the force of friction (F) to the normal force (N).

$\mu = F / N$ or $\mu = \tan(\theta)$ (for inclined plane at angle of repose)

Malayalam support:

[illegible] [illegible] [illegible] [illegible] Friction [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] (Inclined plane) [illegible] [illegible] [illegible] [illegible] [illegible] (Angle) [illegible] Angle of Repose [illegible] [illegible]. [illegible] tangent [illegible] Coefficient of Friction (μ).

Mass Moment of Inertia (Expt 8)

Using a flywheel apparatus, we determine the resistance of a body to rotational motion. This is crucial for designing engines and machinery.

Module IV: Design Thinking

10 Instructional Hours | CO4

Design Thinking Process

This module focuses on user-centric problem solving. Students analyze existing products and propose improvements.

- **Empathize:** Understand user needs (e.g., for a water purifier).
- **Define:** State the problem clearly.
- **Ideate:** Brainstorm creative solutions.
- **Prototype:** Create a simple model or design.
- **Test:** Evaluate the solution.

Case Study: Sunbird Straw (Sustainable product analysis).

Student Activities for Self-Learning

Week	Activity Description	Hours
1-3	Graphical determination of resultants and Jib crane analysis.	4.5
4-5	Link polygon construction for beam reactions.	3.0
6-10	Calculation of Centroids and Friction coefficients.	7.5
11-15	Design thinking practice and final group project.	7.5

Formula & Procedure Bank

Resultant (Analytical)

$$R = \sqrt{(\Sigma F_x^2 + \Sigma F_y^2)}$$
$$\theta = \tan^{-1}(\Sigma F_y / \Sigma F_x)$$

Lami's Theorem

$$P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$$

Friction

$F = \mu N$

$\mu = \tan \theta$

Moment

$M = \text{Force} \times \text{Perpendicular Distance}$

Visual Library

Figure 1: Polygon Law Vector Diagram
Figure 2: Beam Loading Diagram
Figure 3: Friction on Inclined Plane

Assessment Methodology

CIE Breakdown (60 Marks Total)

Component	Marks	Details
Attendance	5	Based on presence in lab sessions
Continuous Evaluation (CE)	30	Record, performance, viva (Every lab)
Practical Test 1 (CA1)	15	First half of experiments (7th week)
Practical Test 2 (CA2)	15	Second half of experiments (14th week)
Open Ended Experiment (OEE)	10	Originality, execution, results (15th week)

ESE (End Semester Exam) - 40 Marks

- Write-up/Procedure: 10 Marks
- Setup & Execution: 10 Marks
- Observation & Result: 8 Marks
- Viva Voce: 8 Marks

- Record: 4 Marks

Module-wise Viva Questions

What is the difference between a Scalar and a Vector?

A scalar has only magnitude (e.g., mass), while a vector has both magnitude and direction (e.g., force).

State the condition for equilibrium of concurrent forces.

The algebraic sum of horizontal components (ΣFx) and vertical components (ΣFy) must both be zero.

What is a 'Link Polygon' used for?

It is a graphical method used to find the magnitude and position of the resultant of non-concurrent forces, or to find support reactions of a beam.

Define 'Angle of Repose'.

It is the maximum angle of an inclined plane at which a body placed on it just begins to slide down due to its own weight.

Practice Model Practical Paper

Time: 3 Hours | Max Marks: 40

1. (a) Determine the resultant of the given concurrent force system using a Force Table. Verify the result graphically. (15 Marks)

OR

(b) Set up a simply supported beam with two unequal point loads. Find the support reactions experimentally and verify using analytical moments. (15 Marks)

2. Determine the coefficient of friction between the given wooden block and the horizontal glass surface. (10 Marks)

3. Viva Voce based on the experiments performed. (8 Marks)

4. Practical Record submission. (7 Marks)

*Note: This is a practice structure based on ESE marks distribution.

Quick Revision & Checklist

Key Points

- Resultant is the single force that produces the same effect as a system of forces.
- Lami's theorem only applies to 3 forces in equilibrium.
- Centroid is the geometric center; Center of Gravity is where weight acts.
- Static friction is always greater than kinetic friction.

Exam Checklist

- [ ] Carry a sharp pencil, scale, and protractor.
- [ ] Check the leveling of the force table using a spirit level.
- [ ] Ensure the string is passing exactly over the center of pulleys.
- [ ] Double-check units (Newton, mm, kg-m^2).

Glossary

Equilibrant
: A force equal and opposite to the resultant that brings the system to rest.

Coplanar Forces
: Forces acting in the same plane.

Moment of Inertia

The property of a body to resist changes to its rotational motion.

Jib

The projecting arm of a crane.

References

- *Engineering Mechanics* by Ramamrutham (S. Chand & Co.)
- *A Text Book of Engineering Mechanics* by Bansal R.K. (Laxmi Publications)
- *Design Thinking: A Beginners Perspective* by E. Balaguruswamy
- NPTEL Course: Engineering Mechanics (IITK)
- Virtual Labs:
